

TBA4245 GEODESY, Assignment no. 2:**Network reliability analysis****Solution****Tasks:**

1. A gross error test in a levelling network.
2. Internal and external reliability analysis of the network.
3. Write computer code to perform the tasks.

See the solution calculated by GISLINE, printouts with estimated gross errors (“Est. grovfeil”) and reliability analysis in the three last pages.

The height difference between a known benchmark A with height 0.000m and an unknown point B are levelled four times. For the second levelling, more accurate equipment was used.

Observations (unit mm) and weights are shown in the ***l***-vector and the ***P***-matrix:



$$l = \begin{bmatrix} 100100 \\ 100099 \\ 100101 \\ 100111 \end{bmatrix} \quad P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- 1) Calculate the weighted average $\hat{l} = \frac{\sum p_i l_i}{\sum p_i}$, the corrections $v_i = \hat{l} - l_i$, the matrices $v = Ax - f$, and $\sum p_i v_i^2 = v^T P v$

Here a table is used; it is recommended to use matrices:

Obs. no.	Observation, l , in mm	Corrections, $v_i = \hat{l} - l_i$	p	$p_i v_i^2$
1	100100	$100102 - 100100 = +2$	1	4
2	100099	+3	2	18
3	100101	+1	1	1
4	100111	-9	1	81
	$\hat{l} = 100102$			104

Matrices:

$$v_i = \hat{l} - l_i \Rightarrow \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \cdot (100102) - \begin{pmatrix} 100100 \\ 100099 \\ 100101 \\ 100111 \end{pmatrix} = \begin{pmatrix} +2 \\ +3 \\ +1 \\ -9 \end{pmatrix}$$

$$\sum p v^2 = \mathbf{v}^T \mathbf{P} \mathbf{v} = 104 \text{ mm}^2$$

2) From 1) we have $\mathbf{A} = [1 \ 1 \ 1 \ 1]^T$ and $\mathbf{f} = l$. Find the Least Squares estimate:

$\hat{\mathbf{x}} = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \mathbf{f}$ (matrix form). Comment on your result.

The observation equations: $\mathbf{v} = \mathbf{A}\mathbf{x} - \mathbf{f}$. The elements in the $\mathbf{A}$ matrix:

For observation no. 1 (of 4): $v_1 = \hat{l} - l_1 \quad v_1 = 1 \cdot \hat{l} - l_1 \quad v_1 = a_1 \cdot \hat{l} - l_1$
 $a_1 = 1 = a_2 = a_3 = a_4$

$$(\hat{l}) = \left[(1 \ 1 \ 1 \ 1) \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right]^{-1} \cdot (1 \ 1 \ 1 \ 1) \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 100100 \\ 100099 \\ 100101 \\ 100111 \end{pmatrix} = (100102)$$

Comment: Same result as in 1), as expected because weighted mean value (weighted mean average) is a least square estimate.

3) Calculate “delete-one” estimates of gross errors in each observation with methods:

- 1) $\nabla_i = l_i - \frac{\sum p' l'}{\sum p'}$ where p' and l' are all the l and p , except l_i and p_i .
- 2) $\nabla_i = l_i - \hat{\mathbf{x}}_i$ where $\hat{\mathbf{x}}_i = (\mathbf{A}^T \mathbf{P}_i \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P}_i \mathbf{f}$ are computed setting a very low weight on observation no. i . For example, $p_i = 0.000001$.
- 3) In part 2: Blunder estimation in the note from *Jon Glenn Gjevestad* (NMBU)

Method 1):

Here, we have to calculate 4 new weighted means, where only $n - 1 = 3$ observations are included: The gross error on an observation is equal to the actual observation no. i subtracted the new weighted mean value where the actual observation is not included.

$$\nabla_1 = l_1 - \frac{\sum p' l'}{\sum p'} = 100100 - \frac{100099 \cdot 1 + 100101 \cdot 1 + 100111 \cdot 1}{2 + 1 + 1} = -2,5 \text{ mm}$$

Same procedure for the others, results:

$$\hat{l}_2 = 100104,0 \quad \hat{l}_3 = 100102,25 \quad \hat{l}_4 = 100099,75 \text{ mm}$$

$$\nabla = \begin{pmatrix} 100100 \\ 100099 \\ 100101 \\ 100111 \end{pmatrix} - \begin{pmatrix} 100102,5 \\ 100104 \\ 100102,25 \\ 100099,75 \end{pmatrix} = \begin{pmatrix} -2,5 \\ -5,0 \\ -1,25 \\ +11,25 \end{pmatrix}$$

Notice that the corrections, v , have the opposite sign of ∇ , the correction is closing the error.

Method 2): In this special case, direct measurements (Ghilani section 1.2) are made in order to estimate the only unknown parameter, the distance itself. Then $\hat{\mathbf{x}}_i$ become a scalar. Weight matrices for $i=1$ to 4:

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \Rightarrow \mathbf{P}_1 = \begin{bmatrix} 0,000001 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{P}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0,000001 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \dots \text{etc}$$

$$i=1: \quad \hat{\mathbf{x}}_1 = (\mathbf{A}^T \mathbf{P}_1 \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P}_1 \mathbf{f} = (4)^{-1} \cdot 400410 = 100102,5 \text{ mm}$$

$$i=2: \quad \hat{\mathbf{x}}_2 = (\mathbf{A}^T \mathbf{P}_2 \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P}_2 \mathbf{f} = (3)^{-1} \cdot 300312 = 100104 \text{ mm}$$

:

and then:

$$\nabla = \begin{pmatrix} 100100 \\ 100099 \\ 100101 \\ 100111 \end{pmatrix} - \begin{pmatrix} 100102,5 \\ 100104 \\ 100102,25 \\ 100099,75 \end{pmatrix} = \begin{pmatrix} -2,5 \\ -5,0 \\ -1,25 \\ +11,25 \end{pmatrix}$$

Conclusion:

The methods 1, 2 (and 3) give identical results. The second method making use of a numerical trick in order to simplify under software developments. In parameter estimation it is also known as the “danish method”.

Method 3), using matrices, see the note from Gjevestad:

Adding an extra column vector with zero elements, except for value 1 for element no. i of the design matrix, $\mathbf{A}$, a blunder can be estimated simultaneously with unknown parameters.

$$\mathbf{A}_i = [\mathbf{A} \quad \mathbf{e}_i]$$

Index i means that we are testing for a blunder in observation i .

Example, $\mathbf{e}_1^T = [1 \quad 0 \quad 0 \quad 0]$ is used to estimate a blunder in observation no. 1:

$$\mathbf{A}_1 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$$

The blunder ∇_1 is estimated simultaneously with the unknown parameters as:

$$\hat{\mathbf{x}}_i = (\mathbf{A}_i^T \mathbf{P} \mathbf{A}_i)^{-1} \mathbf{A}_i^T \mathbf{P} \mathbf{l} \quad \text{where} \quad \hat{\mathbf{x}}_1^T = (\hat{l}_1 \quad \nabla_1)$$

$$\begin{pmatrix} \hat{l}_1 \\ \nabla_1 \end{pmatrix} = \left[\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \right]^{-1} \cdot \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 100100 \\ 100099 \\ 100101 \\ 100111 \end{pmatrix} = \begin{pmatrix} 102,5 \\ -2,5 \end{pmatrix}$$

Same result as in method no. 1.

Same procedure for the other 3 observations. Results:

$$l_2 = 100104 \quad \nabla_2 = -5 \quad l_3 = 100102,25 \quad \nabla_3 = -1,25 \quad l_4 = 100099,75 \quad \nabla_4 = 11,25$$

Compare the results and explain which of the three methods you will prefer.

When using Matlab (or Excel) method no. 3 is better to use compared to method no. 1.

4) Calculate the redundancy matrix of the observations as: $R = I - A(A^T P A)^{-1} A^T P$

For interpretation of the redundancy matrix, see the textbook covering gross errors and internal reliability. Redundancy for the 4 observations, the formula for the R matrix is found in the textbook

$$R = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \left[(1 \quad 1 \quad 1 \quad 1) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right]^{-1} (1 \quad 1 \quad 1 \quad 1) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$R = \begin{pmatrix} 0,8 & -0,4 & -0,2 & -0,2 \\ -0,2 & 0,6 & -0,2 & -0,2 \\ -0,2 & -0,4 & 0,8 & -0,2 \\ -0,2 & -0,4 & -0,2 & 0,8 \end{pmatrix}$$

Notice that observation no. 2 that has the largest weight ($p = 2$), has the lowest value in the main diagonal in the **R**-matrix.

Comments:

The **R**-matrix is generally not diagonal. That means a specific blunder will affect all estimated corrections. It is common to simplify the case, extracting diagonal elements only.

5) Calculate the co-factor matrix of the observation corrections from: $Q_{vv} = RP^{-1}$

Still in this somewhat simplified uncorrelated case:

Observation no. i: $q_{vv} = \frac{r_{ii}}{p_{ii}}$ (independent observations)

The right-hand side consists of known or calculated values. The co-factor matrix of the observation corrections v , only the elements of the main diagonal are of interest here:

$$Q_{vv} = RP^{-1} = \begin{pmatrix} q_{11} & q_{12} & q_{13} & q_{14} \\ q_{21} & q_{22} & q_{23} & q_{24} \\ q_{31} & q_{32} & q_{33} & q_{34} \\ q_{41} & q_{42} & q_{43} & q_{44} \end{pmatrix} = \begin{pmatrix} 0,8 & -0,2 & -0,2 & -0,2 \\ -0,2 & 0,6/2 = 0,3 & -0,2 & -0,2 \\ -0,2 & -0,2 & 0,8 & -0,2 \\ -0,2 & -0,2 & -0,2 & 0,8 \end{pmatrix}$$

6) Calculate the 4 co-factors q_{vv} of the 4 gross errors ∇_i

Must use the Law of Error Propagation (LEP):

Observation no. i: $q_{vv} = \frac{r_{ii}}{p_{ii}}$ (independent observations)

Given:

1) Mathematical linear relations between two random vectors **X** and **Y**: $\mathbf{X} = \mathbf{D}\mathbf{Y} + \mathbf{E}$

2) The covariance matrix $\mathbf{C}_{yy}$ of **Y**

Then the covariance matrix of **X** can be derived as: $\mathbf{C}_{xx} = \mathbf{D}\mathbf{C}_{yy}\mathbf{D}^T$

Here:

1): Mathematical approximation: $\nabla \approx \frac{-1}{r} v = \frac{-1}{p q_{vv}} v$ See also Leick 2004 eq. (4.366), p 157

2): By using LEP we find: $q_{\nabla\nabla} = \left(\frac{-1}{q_{vv}p} \right) q_{vv} \left(\frac{-1}{q_{vv}p} \right) = \frac{1}{q_{vv}p^2}$

Numerical example, observation no. 1: $q_{\nabla 1 \nabla 1} = \frac{1}{q_{vv}p^2} = \frac{1}{0,8 \cdot 1^2} = 1,25$

Instead of following the single element calculations shown above, it is recommended to use the matrix-based method described in the note "Multiple T-test" by Jon Glenn Gjevestad. You get identical numerical results. For matrix **A**_i, see page 4.

The co-factor matrix of the unknown parameters: $\mathbf{Q}_{xx} = (\mathbf{A}_i^T \mathbf{P} \mathbf{A}_i)^{-1} = \begin{bmatrix} - & - & \cdot & - \\ - & - & \cdot & - \\ \cdot & \cdot & \cdot & \cdot \\ - & - & \cdot & q_{vv} \end{bmatrix}$

with the blunders co-factor as the last element on the diagonal. The matrix $(\mathbf{A}_i^T \mathbf{P} \mathbf{A}_i)^{-1}$ is already calculated as part of the estimation of the gross error in method no. 3 on page 4.

Here, for observation no. 1, same result as above, 1,25:

$$\begin{pmatrix} q_{v11} \end{pmatrix} = \begin{bmatrix} (1 & 1 & 1 & 1) \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{pmatrix} \end{bmatrix}^{-1} = \begin{pmatrix} 0,25 & -0,25 \\ -0,25 & 1,25 \end{pmatrix}$$

Same procedure for the other 3 observations.

Numerical values of the elements on the main diagonal only:

$$\mathbf{Q}_{vv} = \begin{pmatrix} q_{v11} & & & \\ & q_{v22} & & \\ & & q_{v33} & \\ & & & q_{v44} \end{pmatrix} = \begin{pmatrix} 1,25 & & & \\ & 0,833 & & \\ & & 1,25 & \\ & & & 1,25 \end{pmatrix}$$

7) Estimate the standard deviations of unit weight ($p = 1$) in the 4 cases.

Remember that the calculation of the gross error on an observation was done by excluding the actual observation from the calculation.

Method no. 1: Starting point is the $\Sigma p v^2 = 104$ in point 2 above, it applies to all 4 observations, including gross error. The following formula can be used, where the minus term in the parentheses is the contribution from the observation with the actual gross error to $\Sigma p v^2$ (is extracted):

$$\hat{\sigma}_{0i}^2 = \frac{1}{f-1} \cdot \left(\sum_1^4 p v v - \frac{v_i^2}{q_{vvii}} \right) = \frac{1}{3-1} \cdot \left(104 - \frac{v_i^2}{q_{vvii}} \right)$$

Observation no. 1:

$$\hat{\sigma}_{01}^2 = \frac{1}{f-1} \cdot \left(\sum_1^4 p v v - \frac{v_1^2}{q_{vv11}} \right) = \frac{1}{3-1} \cdot \left(104 - \frac{(-2,5)^2}{0,8} \right) = 49,50 \quad , \hat{\sigma}_{01} = 7,036 \text{ mm}$$

Method no. 2 (Gjevestad): A posteriori variance for the weight unit, without influence from the blunder ∇_i is estimated (for each i) as:

$$\hat{\sigma}_0^2 = \frac{\mathbf{v}_i^T \mathbf{P} \mathbf{v}_i}{(n - e - 1)}$$

The weighted mean value of 3 observations involved are calculated in point 2, method 3. The 3 estimated corrections, v , on the 3 observations are calculated from: $\hat{v}_i = \hat{l} - l_i$

Example, observation no. 1: using obs. 2, 3 and 4, leaving no. 1 out.

Observations involved: l_2, l_3 and l_4 , see page 1 for numerical values.

Weighted mean value, see page 2 (method 1): $\hat{l}_1 = 100102,5 \text{ mm}$

Corrections: $\hat{v}_2 = \hat{l} - l_2 = 100102,5 - 100099 = 3,5$ $\hat{v}_3 = \hat{l} - l_3 = 1,5$ $\hat{v}_4 = \hat{l} - l_4 = -8,5$

Weights from page 1: $p_2 = 2$ $p_3 = 1$ $p_4 = 1$

$$\text{Matrices: } \mathbf{v}^T \mathbf{P} \mathbf{v} = \begin{pmatrix} 3,5 & 1,5 & -8,5 \end{pmatrix} \cdot \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 3,5 \\ 1,5 \\ -8,5 \end{pmatrix} = (99)$$

$$\hat{\sigma}_0^2 = \frac{\mathbf{v}^T \mathbf{P} \mathbf{v}}{(n - e - 1)} = \frac{99}{(4 - 1 - 1)} = 49,5 \quad \hat{\sigma}_0 = 7,036 \text{ mm, same as by using method no. 1.}$$

The originally number of excess measurements, the degree of freedom: $f = n - e = 4 - 1 = 3$. One observation is removed from the estimations, therefore $f = 2$ and not 3.

Note the loss of one degree of freedom because one observation has been used to estimate the blunder. (Trond's comment: ensures "delete-one" estimates. A blunder in obs. i is adopted in the model sum of squares, and do not contribute to the SSE that is used to estimate $\hat{\sigma}_0^2$).

Same procedure for the other observations, results:

Obs. no. 2: $\hat{\sigma}_0 = 6,083$ mm Obs. no. 3: $\hat{\sigma}_0 = 7,168$ mm Obs. no. 4: $\hat{\sigma}_0 = 1,173$ mm

8) Estimate the standard deviations for the 4 gross errors.

The 4 standard deviations of the gross error with the following known formula: $\hat{\sigma}_v = \hat{\sigma}_0 \cdot \sqrt{q_{vv}}$

Using matrices, the variance/covariance matrix of the gross errors: $C_{vv} = \hat{\sigma}_0^2 \cdot Q_{vv}$

Numerical values of the elements on the main diagonal only:

$$C_{vv} = \begin{pmatrix} \hat{\sigma}_{v1}^2 & & & \\ & \hat{\sigma}_{v1}^2 & & \\ & & \hat{\sigma}_{v1}^2 & \\ & & & \hat{\sigma}_{v1}^2 \end{pmatrix} = \begin{pmatrix} 7,036^2 \cdot 1,25 = 7,87^2 & & & \\ & 5,55^2 & & \\ & & 8,02^2 & \\ & & & 1,31^2 \end{pmatrix}$$

The standard deviations, see the matrix above: 7,87 5,55 8,02 1,31, in mm.

9) Calculate the 4 test statistics values: $t_i = \frac{\hat{v}_i}{\hat{\sigma}_{v_i}}$

All the parameters that are included in the gross error test are now calculated. The test values of the 4 observations, multiple-T-test:

$$t_i = \frac{\hat{v}_i}{\hat{\sigma}_{v_i}} \quad t = \begin{pmatrix} t_1 \\ t_2 \\ t_3 \\ t_4 \end{pmatrix} = \begin{pmatrix} \frac{-2,5}{7,87} \\ \frac{-5,0}{5,55} \\ \frac{-1,25}{8,02} \\ \frac{11,25}{1,31} \end{pmatrix} = \begin{pmatrix} -0,32 \\ -0,90 \\ -0,16 \\ +8,58 \end{pmatrix}$$

Summary:

Answers on most of the questions, denomination is mm:

		2	3	4	5	6	7	8	9
No.	p	v	∇	r	q_{vv}	$q_{\nabla\nabla}$	σ_0	σ_v	t
1	1	+2	-2,5	0,8	0,8	1,25	7,036	7,87	-0,32
2	2	+3	-5,0	0,6	0,3	0,833	6,083	5,55	-0,90
3	1	+1	-1,25	0,8	0,8	1,25	7,169	8,02	-0,16
4	1	-9	+11,25	0,8	0,8	1,25	1,173	1,31	+8,58

10) Compare the test values with your chosen Student-T table value.

Choose a significance level of $\alpha = 5\%$ for the total test.

Must select/calculate the confidence level (test size) of the test of the 4 observations. NB: The actual gross error must be included when the unknowns are counted!

Table value in the test: $T_{\alpha/2, f}$, a two tailed test

Very few observations (< 50): We must calculate the confidence level for the test of 1 observation, α . The starting point is the total confidence level for all 4 observations, $\alpha_{TOT} = 0,05 = 5\%$ (common choice).

Formula from the textbook: $\alpha = 1 - (1 - \alpha_{TOT})^{1/n} = 0,0127 = 1,27\%$

And the degree of freedom is: $f = n - e - 1$ (the gross error) $= 4 - 1 - 1 = 2$ (excess measurements).

Table value: $T_{0,00635, 2} = 8,774$ (0,635 %, two tailed test). Same result as in GISLINE.

11) Make your conclusions: Did you find any gross error?

All test values in column no. 9 in the table above are less than the table value of 8,77.

The null hypothesis (no gross errors) cannot be rejected. The test found no gross error. Observation no. 4 was close (8,58 compared to 8,68) but cannot automatically be characterized as a gross error.

The cause of the unclear conclusion is that there are few observations, making the degree of freedom low, $f = 2$ for the test. Therefore, the standard deviation of the standard deviations in the table above are large; the uncertainties of the standard deviations will be relatively high (large uncertainty).

5 observations: $l_5 = 100,1020$, $p = 1$, Mean value: 100,1020

With 5 instead of 4 observations we have $\alpha = 5\%$ and $f = 5 - 1 - 1 = 3$ (1 extra observation).

The table value is $T_{0,005, 3} = 5,84$, and no. 4 stands out as a gross error (if observation no. 5 has a value equal to the mean value of the 4 first observations), 100102 mm.

		2	3	4	5	6	7	8	9
No.	p	v	∇	R	q_{vv}	$q_{\nabla\nabla}$	σ_0	σ_{∇}	t
1	1	+2,0	-2,4	0,833	0,833	1,20	5,750	6,299	-0,381
2	2	+3,0	-4,5	0,667	0,333	0,75	5,066	4,387	-1,026
3	1	+1,0	-1,20	0,833	0,833	1,20	5,854	6,412	-0,187
4	1	-9,0	+10,80	0,833	0,833	1,20	1,506	1,649	6,548
5	1	+0,0	+0,0	0,833	0,833	1,20	5,888	6,450	0,000

Sum of $p_{vv} = 104$. The degree of freedom (excess measurements): $n - e - 1 = 5 - 1 - 1 = 3$

$\alpha_{tot} = 5\%$ gives $\alpha = 1,02\% = 0,0102$

Table value $T_{(3, 0,05)} = 5,84$ **The test value of obs. no. 4 is 6,548 => Gross error**

12) Interpret the redundancy matrix you calculated in 4.

The redundancy matrix, $\mathbf{R}$, was calculated in 4, the gross error analysis:

$$\mathbf{R} = \begin{pmatrix} 0.8 & -0.4 & -0.2 & -0.2 \\ -0.2 & 0.6 & -0.2 & -0.2 \\ -0.2 & -0.4 & 0.8 & -0.2 \\ -0.2 & -0.4 & -0.2 & 0.8 \end{pmatrix}$$

A redundancy number in an uncorrelated case is defined as $r = \frac{-v}{\nabla}$.

High values of the redundancy numbers in the main diagonal: The sum of the redundancy numbers is equal to 3.0 which is consistent with the number of excess measurements (the degree of freedom).

Lower values of the redundancy number on observation no. 2 which has the highest weight:

For observation no. 2 a smaller percentage of its gross error is reflected in the adjustment of the observation, v , than the other 3 observations (0.6 compared to 0.8, 60% and 80%).

It is logical, why? In the calculations we rely more on an observation with high weight and the measured value is not allowed to change too much.

13) Estimate the largest remaining gross error.

Computes the confidence interval for the largest remaining gross error for the 4 observations:

$$\left[\hat{V} - \hat{\sigma}_V \cdot T_{\alpha/2, f}, \hat{V} + \hat{\sigma}_V \cdot T_{\alpha/2, f} \right]$$

Table value: $T_{\alpha/2, f} = 4.30$, where the significance level is $\alpha = 5\%$ and the degree of freedom is $f = n - e - 1 = 2$. (Losing one more because of the estimation of the gross error)

We have calculated:

Obs. no.	$\hat{V}_i$	$\hat{\sigma}_{V_i}$
1	-2.5	7.87
2	-5.0	5.55
3	-1.25	8.02
4	+11.25	1.31

Thus, the confidence interval for the largest remaining gross error, the table gives by the lower and upper value of the interval:

No.	Lower	Upper	Largest remaining gross error
1	$-2.5 - 7.87 \cdot 4.30 = \mathbf{-36.3}$	+31.3	-36.3
2	-28.9	+18.9	-28.9
3	-35.7	+33.2	-35.7
4	+5.6	+ 16.9	+16.9

The largest remaining gross errors, V_{0i} , is the one of the upper and lower limit of the confidence interval in the table above, that has the largest absolute value. For instance, the two values of observation no. 1 are -36.3 and +31.3 mm, ergo the largest remaining gross error is -36.3. The largest remaining gross errors are marked with **bold** in the table.

14) Calculate and explain the external reliability in this 1D case (levelling) for the 4 gross errors.

The note «Multiple T-test for quality control» from Jon Glenn Gjevestad (NMBU):

∇_x is the effect at estimated parameters from ∇_{max_i} , where ∇_{max} is a column vector of zeroes,

except for the LRE in the i^{th} element: $\nabla_{max}^T = \begin{bmatrix} 0 & 0 & \dots & \nabla_{max_i} & \dots & 0 \end{bmatrix}$.

External reliability explains the deformation a LRE makes at the unknowns ∇_x (or a function of them). In this 1D case with only one unknown, the only possible deformation to compute is a change in height at point B caused by a LRE in one of the observations.

According to Gjevestad's note:

$$\nabla_x = (\mathbf{A}^T \mathbf{P} \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \nabla_{max}$$

The LRE of obs. no. 3 is used to calculate the deformation on the height of B caused by this LRE:

$$(\nabla_x) = \left[(1 \ 1 \ 1 \ 1) \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right]^{-1} \cdot (1 \ 1 \ 1 \ 1) \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ -35,7 \\ 0 \end{pmatrix} = (-7,14 \text{ mm})$$

Same procedure for the other 3 observations. Results:

$$l_1: \nabla_x = -7.26 \text{ mm} \quad l_2: \nabla_x = -11.56 \text{ mm} \quad l_3: \nabla_x = -7.14 \text{ mm} \quad l_4: \nabla_x = 3.38 \text{ mm}$$

Conclusion:

We have calculated the maximum deformation caused by each of the largest remaining gross error (LRE) on the 4 observations. It is a LRE of -28,9 mm on observation no. 2 that will give the largest deformation on the height of point B, -11,56 mm = -0,0116 m. Notice that we are using the “the worst case” of the 4 as the measure of external reliability.

Logical that obs. no. 2 is making largest deformation of the height, as this observation has weight $p = 2$, the double of the 3 others with $p = 1$, has more weight and thus more influence on the results.

Gisline documentation:

NYBESTEMTE KOORDINATER MED MIDLERE FEIL [meter]. *Estimated coordinates*

PUNKT	X	Y	H	sX	sY	sH
B			100.1020			0.0026

KORRIGERTE OBSERVASJONER, ANTATT MIDLERE FEIL OG RESTFEIL [meter / gon]

Corrected observations. Assumed standard deviation and residuals

Fra From	Til To	Observasjon Observation	Antatt M Assumed SD	Restfeil Residual
A	1 B	dH 100.1000	0.0010	0.0020
A	2 B	dH 100.0990	0.0007	0.0030 *
A	3 B	dH 100.1010	0.0010	0.0010
A	4 B	dH 100.1110	0.0010	-0.0090 ***

STATISTIKK

Antall iterasjoner	:	1	
Antall observasjoner høydeforskjell	:	4	
Antall observasjoner	:	4	No. of obs.
Antall ukjente høydekoordinater	:	1	
Antall ukjente	:	1	No. of unknowns
Antall overbestemmelser	:	3	Degree of freedom
Feilkvadratsum	:	104.00542964	sum PVV
Beregnet std.avvik på vektsenheten	:	5.8880	Estimated SD
Antatt std.avvik på vektsenheten	:	1.0000	Assumed SD of the obs. with weight p = 1

TEST AV OBSERVASJONER - MULTIPPEL T-TEST

Fra From	Til To	Restfeil Residual	Est.grovfeil Est. gross error	Testverdi Test value
A	1 B	dH 0.0020	-0.0025	0.32
A	2 B	dH 0.0030	-0.0050	0.90
A	3 B	dH 0.0010	-0.0012	0.16
A	4 B	dH -0.0090	0.0113	8.58

Tabellverdi=8.77 (Student-t, f=2, alfa=0.0064)

Table value

OPPSUMERING ETTER TEST AV OBSERVASJONER: *Summary, gross error test*

Kategori	Ant.obs.	Akkumulert (%)	
Test/Tabell < 1.0	4	100.00	No gross error
1.0 < Test/Tabell < 2.0	0	100.00	
2.0 < Test/Tabell < 3.0	0	100.00	
3.0 < Test/Tabell < 3.0	0	100.00	
Ukontrollerbar	0	100.00	

STATISTIKK

Antall iterasjoner	:	1
Antall observasjoner høydeforskjell	:	4
Antall observasjoner	:	4
Antall ukjente høydekoordinater	:	2
Antall tilleggsukjente	:	2
Antall ukjente	:	4
Rangdefekt	:	3
Antall ukjente korrigert	:	1
Antall overbestemmelser	:	3
Feilkvadratsum	:	104.00542964
Beregnet std.avvik på vektsenheten	:	5.8880
Antatt std.avvik på vektsenheten	:	1.0000

TEST AV M0

Tabellverdi = 7.82 (Kjikkvadrat, f=3, alfa=0.0500)
 Beregnet verdi = 104.01 ***

Ingen feil i observasjonsmaterialet er funnet
NO GROSS ERRORS FOUND

Printout from GISLINE, reliability of the unknown height of B:

The Norw. land surveying program Gisline gives the same result as in the calculations above.
From the Gisline report file (dok file):

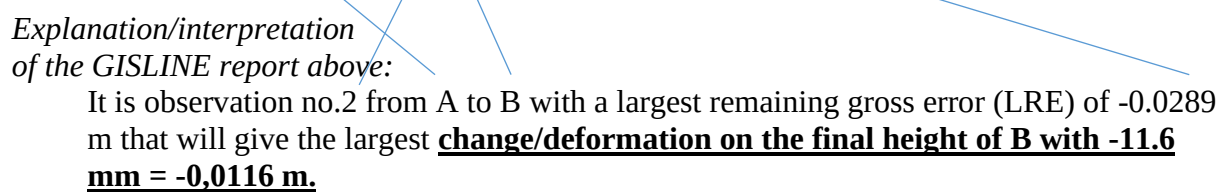
```

YTRE PÅLITELIGHET - KOORDINATER [meter]
External reliability - Coordinates

KOORDINAT Observasjon [meter/gon]      Indre pål.      Ytre pål.
Internal reliab. External

reliab.
-----
--
H B      A      2 B      dH      -0.0289      -0.0116

```



*Explanation/interpretation
of the GISLINE report above:*

It is observation no.2 from A to B with a largest remaining gross error (LRE) of -0.0289 m that will give the largest **change/deformation on the final height of B with -11.6 mm = -0,0116 m.**